Functional Annotation Report: *dfp / coaBC* (CoaBC) in *Pseudomonas putida* KT2440

UniProt: Q88C96 · **Gene:** *dfp* (synonym *coaBC*) · **Locus:** PP_5285 **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440) **EC numbers:** 6.3.2.5 (phosphopantothenate–cysteine ligase) + 4.1.1.36 (phosphopantothenoylcysteine decarboxylase)

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1. Summary (Answer to the Research Question)

Q88C96 (*dfp*/*coaBC*, PP_5285) encodes **CoaBC**, a **bifunctional, cytoplasmic enzyme that carries out the second and third of the five steps of the universal coenzyme A (CoA) biosynthetic pathway**, converting 4′-phosphopantothenate into 4′-phosphopantetheine. It is a single polypeptide with two catalytic domains:

- A **C-terminal CoaB domain** = **phosphopantothenoylcysteine synthetase (PPCS, EC 6.3.2.5)**, which ligates L-cysteine to 4′-phosphopantothenate to form (R)-4′-phospho-*N*-pantothenoylcysteine (PPC), using **CTP** (not ATP) and Mg^{2+}.
- An **N-terminal CoaC domain** = **phosphopantothenoylcysteine decarboxylase (PPCDC, EC 4.1.1.36)**, an **FMN-dependent** member of the HFCD flavoprotein family that decarboxylates PPC to 4′-phosphopantetheine.

CoaBC is **essential**, acts in the **cytosol**, forms a higher-order oligomer (dodecamer) subject to **feedback regulation by CoA thioesters**, and is a validated antibacterial drug target.

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2. Identity Verification

The assignment is unambiguous and consistent across independent lines of evidence:

- **Gene symbol history:** The *dfp* gene ("DNA/pantothenate metabolism flavoprotein") was renamed ***coaBC*** when Strauss et al. showed the Dfp protein carries both CoA-pathway

activities
(P 11278255).
The UniProt record's dual naming (*dfp* / *coaBC*; PPCS-PPCDC; CoaB + CoaC) matches this literature exactly.

- **Domain/family match:** The provided InterPro/family annotation (HFCD flavoprotein in the N-terminal section; CoaB-like_sf; CoaBC; DNA/pantothenate-metab_flavo_C; Flavoprotein) precisely matches the biochemically defined two-domain architecture of Dfp/CoaBC.
- **Organism match:** Pseudomonad-specific CoA-pathway physiology (below) is documented for the closely related *P. aeruginosa*
(P 21551303),
directly relevant to *P. putida*.

No conflicting gene of the same symbol was encountered. The literature for "*dfp*/*coaBC*" describes this exact bifunctional CoA-biosynthesis flavoprotein.

Target-specific sequence evidence (Q88C96, 403 aa): Direct analysis of the *P. putida* KT2440 sequence confirms both intact catalytic domains and their conserved active-site residues: - **N-terminal PPCDC/CoaC domain (~1–190):** carries the HFCD flavoprotein signature **G-G-I-A-A-Y-K at residues 15–21**, matching Kupke's conserved G-G/S-I-A-X-Y-K coenzyme/substrate-binding motif
(P 11358972);
a **Cys159** lies in the region homologous to the *E. coli* Dfp substrate-clamp Cys158. - **C-terminal PPCS/CoaB domain (~200–403):** carries the strictly conserved synthetase motif **K-L-K-K at residues 288–291**, corresponding to the essential *E. coli* Lys289 (KXKK) motif
(P 12140293).
- The N-terminus (MQRLYRKRIV…) is basic/hydrophilic with **no signal peptide or transmembrane segment**, consistent with cytoplasmic localization.

This confirms the orthology-based assignment at the level of the actual *P. putida* protein rather than relying solely on distant homologs.

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3. Primary Function: Reaction Catalyzed and Substrate Specificity

Overall reaction (two consecutive pathway steps):

> 4′-phosphopantothenate + L-cysteine + CTP → (via PPC) → 4′-phosphopantetheine + CO_2 + CMP + PPi (+ Mg^{2+} required)

Strauss et al. identified PPCS as "the last unidentified coenzyme A biosynthetic enzyme in bacteria" and showed that the *dfp* product (renamed *coaBC*) "also has phosphopantothenoylcysteine synthetase activity, using CTP rather than ATP as the activating nucleoside 5′-triphosphate," which "completes the identification of all the enzymes involved in the biosynthesis of coenzyme A in bacteria" (P 11278255).

Balibar et al. describe *coaBC* as "a single gene encoding a bifunctional protein catalyzing two consecutive steps in the CoA pathway converting 4′-phosphopantothenate to 4′-phosphopantetheine" (P 21551303).

3a. CoaB / PPCS domain (EC 6.3.2.5) — C-terminal

- **Substrates:** 4′-phosphopantothenate, L-cysteine, CTP, Mg^{2+}.
- **Substrate specificity for the nucleotide:** distinctively uses **CTP** rather than ATP (P 11278255) — a hallmark that distinguishes bacterial PPCS from the eukaryotic (ATP-using) enzyme.
- **Mechanism:** proceeds in **two half-reactions**. The phosphopantothenate carboxyl is first activated as an **acyl-cytidylate (phosphopantothenoyl-cytidylate) intermediate**, then attacked by the cysteine amine. Kupke trapped this intermediate using mutant His-CoaB N210D (and with wild type when cysteine was omitted), and showed the strictly conserved **^{289}KXKK^{292} motif (Lys289)** is essential — its exchange abolishes both intermediate formation and product (P 12140293).

The CoaB domain forms **dimers**
(P 12140293).

3b. CoaC / PPCDC domain (EC 4.1.1.36) — N-terminal, flavoprotein

- **Substrate/product:** decarboxylates (R)-4′-phospho-*N*-pantothenoylcysteine (PPC) to **4′-phosphopantetheine**
(P 11358972).
- **Cofactor: FMN**, bound **non-covalently at the interface between two subunits** within a Rossmann fold; the domain belongs to the **HFCD** ("homooligomeric flavin-containing Cys decarboxylase") family, together with the lantibiotic LanD enzymes EpiD/MrsD and the plant salt-tolerance protein AtHAL3a
(P 11358972 ,
P 15459342).
Because the flavin sits at a subunit interface, oligomerization (trimer → dodecamer) is required to assemble a functional active site
(P 15459342).
- **Mechanism:** an **FMN-mediated oxidative decarboxylation**. The reaction proceeds through an **oxidatively decarboxylated aminoenethiol intermediate** (loss of CO_2 + two H atoms, confirmed by high-resolution MS), which is then **re-reduced by a redox-active cysteine** (Cys175 in AtHAL3a; equivalents Asn125/Cys158 essential in bacterial Dfp) to yield the saturated phosphopantetheine thiol
(P 11923307 ,
P 11358972).
Key catalytic/substrate-clamp residues in the conserved G-G/S-I-A-X-Y-K motif are required for flavin binding and activity
(P 11358972).

Note on domain order vs. reaction order: In the polypeptide the **decarboxylase (CoaC) is N-terminal** and the **synthetase (CoaB) is C-terminal**, whereas in the pathway the synthetase acts first — hence the name "Coa**B**C."

4. Subcellular Localization

CoaBC is a **soluble cytoplasmic enzyme**. CoA biosynthesis is a cytosolic process, and CoaBC has been characterized as a soluble oligomeric protein (gel filtration; native MS; crystallography of homologs) with no signal peptide or membrane-spanning region in Q88C96 (P 12140293, P 33420031). Its function — producing the CoA precursor 4′-phosphopantetheine — is therefore carried out **inside the cell, in the cytoplasm**, where downstream CoaD/CoaE complete CoA assembly.

5. Pathway Context and Regulation

- **Pathway:** CoA is assembled in **five steps from pantothenate** (PanK/CoaA(X) → **PPCS(CoaB) → PPCDC(CoaC)** → PPAT/CoaD → DPCK/CoaE); the intermediates are common to prokaryotes and eukaryotes (P 15893380). CoaBC occupies steps 2–3, downstream of pantothenate kinase and upstream of phosphopantetheine adenylyltransferase.
- **Downstream fate of the product:** The 4′-phosphopantetheine made by CoaBC is converted by CoaD (PPAT) and CoaE (DPCK) into **free CoA**. CoA is then the donor from which **4′-phosphopantetheinyl transferases (PPTases; AcpS, Sfp)** transfer the 4′-phosphopantetheine moiety onto a conserved serine of apo-**acyl carrier proteins (ACP)** and **peptidyl carrier proteins (PCP)**, producing the holo forms whose phosphopantetheine "swinging arm" thiol is essential for fatty-acid, polyketide, and nonribosomal-peptide synthesis (P 15065855, P 11525165). Thus CoaBC feeds both the central CoA pool and the carrier-protein machinery — the basis of its essentiality.

- **Regulation:** The first full-length CoaBC structure (from *Mycobacterium smegmatis*) shows the enzyme "is organised as a **dodecamer** and **regulated by CoA thioesters**," with inhibitors binding "a **cryptic allosteric site within CoaB**" (P 33420031). This implies end-product/thioester feedback control over CoA output at the CoaBC node.
- **Pseudomonad-specific feature:** In *Pseudomonas*, the upstream pantothenate kinase is the **type III CoaX** (not *E. coli*'s CoaA). Consequently *P. aeruginosa* — unlike *E. coli* — **cannot use pantethine to bypass loss of coaBC**, because of the differing substrate specificity of its pantothenate kinase (P 21551303). This indicates pseudomonads, including *P. putida*, depend on **de novo CoaBC activity** with no salvage bypass at this step.

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6. Essentiality and Biomedical/Biotechnological Relevance

- **Essential gene:** *coaBC* is essential in *E. coli* (arabinose-regulated depletion) and its depletion is **bactericidal** in *M. tuberculosis* (P 21551303, P 33420031).
- **Validated drug target:** The CoA biosynthetic pathway "is a target for antibacterial drug discovery" (P 15893380); CoaBC specifically has been pursued with high-throughput biochemical screens and native ion-mobility MS to find selective inhibitors of the CoaB domain (P 33420031, P 31488574). The same essentiality principle underlies its interest in *Pseudomonas*.
- **Essentiality in *P. putida* KT2440 (inference):** Genome-scale metabolic models of KT2440 predict a defined essential-gene set, and a genome-wide mini-Tn5 knockout screen

recovered only *conditionally* essential auxotrophs (amino acids, biotin, nicotinate, vitamin B12) rescuable by supplementation (P 20158506). Because CoA cannot be imported and this pathway step cannot be salvaged in *Pseudomonas* (no pantethine bypass; P 21551303), *coaBC* (PP_5285) is expected to be **unconditionally essential** — consistent with its being absent from recoverable mutant/auxotroph collections and with demonstrated essentiality of *coaBC* in *E. coli* and *M. tuberculosis*.

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7. Supported vs. Refuted Hypotheses

Supported: - H1: Q88C96 is the bifunctional CoaBC (PPCS + PPCDC) of CoA biosynthesis — **strongly supported** (multiple primary studies of orthologs; family/domain match). - H2: The synthetase uses CTP and an acyl-cytidylate intermediate — **supported** (P 11278255 P 12140293). - H3: The decarboxylase is an FMN-dependent HFCD flavoprotein acting via an aminoenethiol intermediate — **supported** (P 11358972 P 11923307). - H4: The enzyme is cytoplasmic and essential — **supported** (P 33420031 P 21551303).

Refuted / ruled out: - The bacterial synthetase does **not** use ATP as activator (uses CTP) — the ATP hypothesis is refuted for this domain (P 11278255).

- A pantethine-salvage bypass of CoaBC does **not** operate in *Pseudomonas* (refuted for pseudomonads;

P 21551303).

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8. Limitations and Future Directions

- **Direct evidence gap:** Mechanistic/structural data derive from orthologs (*E. coli* Dfp, *M. smegmatis*/*M. tuberculosis* CoaBC, *Arabidopsis* AtHAL3a). No *P. putida* KT2440-specific enzymological study was located; the annotation for Q88C96 is by strong orthology/inference. Direct biochemical confirmation (CTP specificity, FMN content, k_cat/K_m) and a *P. putida* structure would close this gap.
- **Regulation specifics:** Feedback by CoA thioesters is established in mycobacterial CoaBC; whether the *P. putida* enzyme shares the same allosteric site/kinetics remains to be tested.
- **Future work:** Verify essentiality in *P. putida* KT2440 (e.g., conditional knockdown), determine the oligomeric state and FMN stoichiometry, and evaluate CoaB-domain inhibitors for antipseudomonal activity.

References (PMIDs)

- 11278255 — Strauss et al., *J. Biol. Chem.* 2001. PPCS identification; *dfp→coaBC*; CTP-dependent synthetase.
- 12140293 — Kupke, *J. Biol. Chem.* 2002. CoaB domain mechanism; acyl-cytidylate intermediate; Lys289 (KXKK) motif; CoaB dimerization.
- 11358972 — Kupke, *J. Biol. Chem.* 2001. N-terminal PPCDC domain; HFCD flavoprotein family; conserved motifs.
- 11923307 — Hernández-Acosta et al., 2002. AtHAL3a; general PPCDC oxidative-decarboxylation mechanism (aminoenethiol intermediate).
- 21551303 — Balibar et al., 2011. Essentiality of *coaBC*; Pseudomonas pantothenate kinase (CoaX) specificity; no pantethine bypass.
- 33420031 — Mendes et al., 2021. First full-length CoaBC structure (dodecamer); CoA-thioester regulation; allosteric CoaB inhibitors; bactericidal on depletion.

- 31488574 — Chan et al., 2019. Native IM-MS structural analysis of *E. coli* PPCS domain; inhibitor screening.
- 15893380 — Leonardi et al., *Prog. Lipid Res.* 2005. Review: CoA biosynthesis genetics/enzymology; antibacterial target.
- 15459342 — Rangarajan et al., 2004. HFCD/FMN flavoprotein fold; non-covalent FMN at subunit interface; trimer→dodecamer assembly (Pad1, EpiD, MrsD, AtHAL3a).
- 15065855 — Mofid et al., 2004. PPTase (Sfp) mechanism; transfer of 4′-phosphopantetheine from CoA to ACP/PCP serine.
- 11525165 — Xu et al., 2001. ACP structure; 4′-phosphopantetheine prosthetic group / swinging arm derived from CoA.
- 20158506 — Molina-Henares et al., 2010. Genome-wide *P. putida* KT2440 mutant screen; essential/conditionally essential gene identification.
